

STEP 0 — Prerequisites Check

```
Microsoft Windows [Version 10.0.26200.9445]
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C:\Users\Hirensinh\Downloads\Container Orchestration Assignment>docker --version
Docker version 29.3.1, build c2be9cc

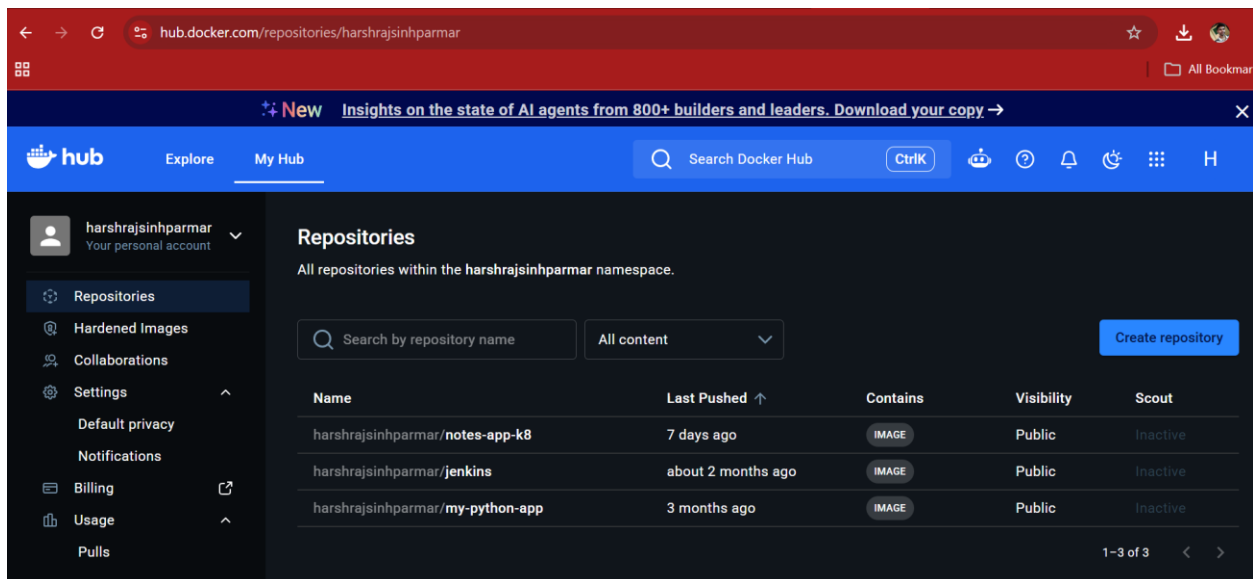
C:\Users\Hirensinh\Downloads\Container Orchestration Assignment>kubectl version --client
Client Version: v1.34.1
Kustomize Version: v5.7.1

C:\Users\Hirensinh\Downloads\Container Orchestration Assignment>helm version
version.BuildInfo{Version:"v3.16.1", GitCommit:"5a5449dc42be07001fd5771d56429132984ab3ab", GitTreeState:"clean", GoVersion:"go1.22.7"}

C:\Users\Hirensinh\Downloads\Container Orchestration Assignment>minikube version
minikube version: v1.37.0
commit: 65318f4cfff9c12cc87ec9eb8f4cdd57b25047f3

C:\Users\Hirensinh\Downloads\Container Orchestration Assignment>git --version
git version 2.53.0.windows.2

C:\Users\Hirensinh\Downloads\Container Orchestration Assignment>
```



The screenshot shows the Docker Hub profile page for the user 'harshrajsinhparmar'. The page has a blue header with the Docker Hub logo and navigation links. The left sidebar contains a list of account settings: Repositories, Hardened Images, Collaborations, Settings, Default privacy, Notifications, Billing, Usage, and Pulls. The main content area is titled 'Repositories' and shows a list of repositories within the user's namespace. The list includes three repositories: 'notes-app-k8', 'jenkins', and 'my-python-app'. Each repository entry shows its name, last pushed date, contains type (IMAGE), visibility (Public), and status (Inactive). A 'Create repository' button is visible in the top right corner of the repositories section.

Name	Last Pushed	Contains	Visibility	Scout
harshrajsinhparmar/notes-app-k8	7 days ago	IMAGE	Public	Inactive
harshrajsinhparmar/jenkins	about 2 months ago	IMAGE	Public	Inactive
harshrajsinhparmar/my-python-app	3 months ago	IMAGE	Public	Inactive

STEP 1 — Start Minikube & Set Variables

```
C:\Users\Hirensinh\Downloads\Container Orchestration Assignment>minikube start --cpus=4 --memory=6144 --driver=docker
* minikube v1.37.0 on Microsoft Windows 11 Pro 10.0.26200.9445 Build 26200.9445
* minikube 1.39.0 is available! Download it: https://github.com/kubernetes/minikube/releases/tag/v1.39.0
* To disable this notice, run: 'minikube config set WantUpdateNotification false'

* Using the docker driver based on existing profile

X Exiting due to MK_USAGE: Docker Desktop has only 3769MB memory but you specified 6144MB

C:\Users\Hirensinh\Downloads\Container Orchestration Assignment>minikube delete
* Deleting "minikube" in docker ...
* Removing C:\Users\Hirensinh\.minikube\machines\minikube ...
* Removed all traces of the "minikube" cluster.
```

```
C:\Users\Hirensinh\Downloads\Container Orchestration Assignment>minikube start --cpus=2 --memory=3072 --driver=docker
* minikube v1.37.0 on Microsoft Windows 11 Pro 10.0.26200.9445 Build 26200.9445
* Using the docker driver based on user configuration
* Using Docker Desktop driver with root privileges
* Starting "minikube" primary control-plane node in "minikube" cluster
* Pulling base image v0.0.48 ...
* Creating docker container (CPUs=2, Memory=3072MB) ...
! Failing to connect to https://registry.k8s.io/ from inside the minikube container
* To pull new external images, you may need to configure a proxy: https://minikube.sigs.k8s.io/docs/reference/networking/proxy/
* Preparing Kubernetes v1.34.0 on Docker 28.4.0 ...
* Configuring bridge CNI (Container Networking Interface) ...
* Verifying Kubernetes components...
  - Using image gcr.io/k8s-minikube/storage-provisioner:v5
* Enabled addons: default-storageclass, storage-provisioner
* Done! kubectl is now configured to use "minikube" cluster and "default" namespace by default
```

Enable Ingress addon

```
C:\Users\Hirensinh\Downloads\Container Orchestration Assignment>minikube addons enable ingress
* ingress is an addon maintained by Kubernetes. For any concerns contact minikube on GitHub.
You can view the list of minikube maintainers at: https://github.com/kubernetes/minikube/blob/master/OWNERS
* After the addon is enabled, please run "minikube tunnel" and your ingress resources would be available at "127.0.0.1"
  - Using image registry.k8s.io/ingress-nginx/kube-webhook-certgen:v1.6.2
  - Using image registry.k8s.io/ingress-nginx/controller:v1.13.2
  - Using image registry.k8s.io/ingress-nginx/kube-webhook-certgen:v1.6.2
* Verifying ingress addon...
* The 'ingress' addon is enabled
```

Clone the Repository & Go Inside

```
C:\Users\Hirensinh\Downloads\Container Orchestration Assignment>$env:DOCKERHUB_USERNAME="harshrajsinhparmar"
The filename, directory name, or volume label syntax is incorrect.

C:\Users\Hirensinh\Downloads\Container Orchestration Assignment>git clone https://github.com/UnpredictablePrashant/StreamingApp.git
Cloning into 'StreamingApp'...
remote: Enumerating objects: 479, done.
remote: Counting objects: 100% (128/128), done.
remote: Compressing objects: 100% (65/65), done.
remote: Total 479 (delta 90), reused 63 (delta 63), pack-reused 351 (from 1)
Receiving objects: 100% (479/479), 15.98 MiB | 2.27 MiB/s, done.
Resolving deltas: 100% (157/157), done.

C:\Users\Hirensinh\Downloads\Container Orchestration Assignment>cd StreamingApp

C:\Users\Hirensinh\Downloads\Container Orchestration Assignment\StreamingApp>
```

Login to Docker Hub

```
C:\Users\Hirensinh\Downloads\Container Orchestration Assignment\StreamingApp>docker login
Authenticating with existing credentials... [Username: harshrajsinhparmar]

Info → To login with a different account, run 'docker logout' followed by 'docker login'

Login Succeeded
```

Build All 5 Docker Images

1)

```

C:\Users\Hirensinh\Downloads\Container Orchestration Assignment\StreamingApp>docker build -t harshrajsinhparmar/streamin
g-auth:1.0.0 ./backend/authService
[+] Building 96.5s (11/11) FINISHED                                docker:desktop-linux
=> [internal] load build definition from Dockerfile                1.5s
=> => transferring dockerfile: 222B                                0.6s
=> [internal] load metadata for docker.io/library/node:18-alpine   6.8s
=> [auth] library/node:pull token for registry-1.docker.io         0.0s
=> [internal] load .dockerignore                                    0.2s
=> => transferring context: 98B                                      0.1s
=> [1/5] FROM docker.io/library/node:18-alpine@sha256:8d6421d663b4c28fd3ebc498332f249011d118945588d0a35cb9bc4b8 29.2s
=> => resolve docker.io/library/node:18-alpine@sha256:8d6421d663b4c28fd3ebc498332f249011d118945588d0a35cb9bc4b8c 0.2s
=> => sha256:1e5a4c89cee5c0826c540ab06d4b6b491c96eda01837f430bd47f0d26702d6e3 1.26MB / 1.26MB 5.0s
=> => sha256:f18232174bc91741fd3da96d85011092101a032a93a388b79e99e69c2d5c870 3.64MB / 3.64MB 8.4s
=> => sha256:25ff2da83641908f65c3a74d80409d6b1b62ccfaab220b9ea70b80df5a2e0549 446B / 446B 2.0s
=> => sha256:dd71dde834b5c203d162902e6b8994cb2309ae049a0eabc4feea161b2b5a3d0e 40.01MB / 40.01MB 19.0s
=> => extracting sha256:f18232174bc91741fd3da96d85011092101a032a93a388b79e99e69c2d5c870 1.4s
=> => extracting sha256:dd71dde834b5c203d162902e6b8994cb2309ae049a0eabc4feea161b2b5a3d0e 7.1s
=> => extracting sha256:1e5a4c89cee5c0826c540ab06d4b6b491c96eda01837f430bd47f0d26702d6e3 0.3s
=> => extracting sha256:25ff2da83641908f65c3a74d80409d6b1b62ccfaab220b9ea70b80df5a2e0549 0.1s
=> [internal] load build context                                    1.8s
=> => transferring context: 91.62kB                                  1.5s
=> [2/5] WORKDIR /app                                              0.5s
=> [3/5] COPY package*.json ./                                     0.2s
=> [4/5] RUN npm install --production                             34.2s
=> [5/5] COPY . .                                                  0.4s
=> exporting to image                                              18.6s
=> => exporting layers                                              12.6s
=> => exporting manifest sha256:9c656380344f58eb54f57b013733bf3003626048e72e6057259586017e6ac159 0.1s
=> => exporting config sha256:4fcb3b89016a376b778f17c519256f632377e0d0d4a35eacd277bed156a89aee 0.1s
=> => exporting attestation manifest sha256:b9ae2e03169ad228308dc279f355a264b939833e23d3b93805b1384dff9084b4 0.1s

=> => exporting manifest list sha256:532dee24430d7a4200e695677137d7468df448da64ad2631502956725f1cd729 0.0s
=> => naming to docker.io/harshrajsinhparmar/streaming-auth:1.0.0 0.0s
=> => unpacking to docker.io/harshrajsinhparmar/streaming-auth:1.0.0 5.2s

View build details: docker-desktop://dashboard/build/desktop-linux/desktop-linux/5iy5cvud10ze1cvaogc8h98gz

```

2)

```

C:\Users\Hirensinh\Downloads\Container Orchestration Assignment\StreamingApp>docker build -t harshrajsinhparmar/streamin
g-stream:1.0.0 -f ./backend/streamingService/Dockerfile ./backend
[+] Building 87.6s (10/10) FINISHED                                docker:desktop-linux
=> [internal] load build definition from Dockerfile                0.6s
=> => transferring dockerfile: 272B                                0.2s
=> [internal] load metadata for docker.io/library/node:18-alpine   1.4s
=> [internal] load .dockerignore                                    0.1s
=> => transferring context: 2B                                       0.0s
=> [internal] load build context                                    3.6s
=> => transferring context: 15.79MB                                  3.5s
=> CACHED [1/5] FROM docker.io/library/node:18-alpine@sha256:8d6421d663b4c28fd3ebc498332f249011d118945588d0a35cb 0.1s
=> => resolve docker.io/library/node:18-alpine@sha256:8d6421d663b4c28fd3ebc498332f249011d118945588d0a35cb9bc4b8c 0.1s
=> [2/5] WORKDIR /app/streamingService                             0.2s
=> [3/5] COPY streamingService/package*.json ./                    0.3s
=> [4/5] RUN npm install --production                             65.1s
=> [5/5] COPY streamingService/. ./                                1.2s
=> exporting to image                                              14.1s
=> => exporting layers                                              9.6s
=> => exporting manifest sha256:ef24f2acb76826575f930eeb87b7409ec557b34d31710ff2902aceaabb7ed497 0.0s
=> => exporting config sha256:35cf0b4729959ec7cbe4c1a9aed67ea10e5fc303b73c00fe0eb4a2de1bdf8ed6 0.0s
=> => exporting attestation manifest sha256:0bbcc6c0d395ff26dd5b3d8d3ce1c97cde5b815186d63760757b9f0f3ac908f75 0.0s
=> => exporting manifest list sha256:d3dabb86d7d58ac5dbb18423b15c2d267580ddd47d759446d5eb2c6def10135d 0.0s
=> => naming to docker.io/harshrajsinhparmar/streaming-stream:1.0.0 0.0s
=> => unpacking to docker.io/harshrajsinhparmar/streaming-stream:1.0.0 4.2s

View build details: docker-desktop://dashboard/build/desktop-linux/desktop-linux/ze4exj3ogm567hkoj8si557r8

```

3)

```
C:\Users\Hirensinh\Downloads\Container Orchestration Assignment\StreamingApp>docker build -t harshrajsinhparmar/streaming-admin:1.0.0 -f ./backend/adminService/Dockerfile ./backend
[+] Building 54.1s (11/11) FINISHED                                docker:desktop-linux
=> [internal] load build definition from Dockerfile                0.1s
=> => transferring dockerfile: 260B                                0.1s
=> [internal] load metadata for docker.io/library/node:18-alpine  1.8s
=> [auth] library/node:pull token for registry-1.docker.io        0.0s
=> [internal] load .dockerignore                                  0.0s
=> => transferring context: 2B                                       0.0s
=> CACHED [1/5] FROM docker.io/library/node:18-alpine@sha256:8d6421d663b4c28fd3ebc498332f249011d118945588d0a35cb 0.2s
=> => resolve docker.io/library/node:18-alpine@sha256:8d6421d663b4c28fd3ebc498332f249011d118945588d0a35cb9bc4b8c 0.1s
=> [internal] load build context                                  0.5s
=> => transferring context: 153.90kB                                0.5s
=> [2/5] WORKDIR /app/adminService                               0.2s
=> [3/5] COPY adminService/package*.json ./                      0.3s
=> [4/5] RUN npm install --production                           38.8s
=> [5/5] COPY adminService/. ./                                  0.5s
=> exporting to image                                           11.2s
=> => exporting layers                                              7.0s
=> => exporting manifest sha256:137b8eb19a91f3bdd59cf66e132023dd0b2d21599cf13c4236a7889b49705ba6 0.0s
=> => exporting config sha256:2c83405fa0263056a2e3442dfb0e3c9a43e90025c359a5c7cea5df8d8e8450b3 0.0s
=> => exporting attestation manifest sha256:460a92ad9f227ca4713a51faad181659c7d76f6755ce210add1dc9ddaba5a079 0.0s
=> => exporting manifest list sha256:45be2235ac996292eaff3e3440d513e106aa529024526021a47be1ebaeabf3f5 0.0s
=> => naming to docker.io/harshrajsinhparmar/streaming-admin:1.0.0 0.0s
=> => unpacking to docker.io/harshrajsinhparmar/streaming-admin:1.0.0 3.9s

View build details: docker-desktop://dashboard/build/desktop-linux/desktop-linux/9x0u931d69na0yhd58u0ag6kg
```

4)

```
C:\Users\Hirensinh\Downloads\Container Orchestration Assignment\StreamingApp>docker build -t harshrajsinhparmar/streaming-chat:1.0.0 -f ./backend/chatService/Dockerfile ./backend
[+] Building 22.6s (10/10) FINISHED                                docker:desktop-linux
=> [internal] load build definition from Dockerfile                0.2s
=> => transferring dockerfile: 257B                                0.1s
=> [internal] load metadata for docker.io/library/node:18-alpine  1.3s
=> [internal] load .dockerignore                                  0.0s
=> => transferring context: 2B                                       0.0s
=> CACHED [1/5] FROM docker.io/library/node:18-alpine@sha256:8d6421d663b4c28fd3ebc498332f249011d118945588d0a35cb 0.1s
=> => resolve docker.io/library/node:18-alpine@sha256:8d6421d663b4c28fd3ebc498332f249011d118945588d0a35cb9bc4b8c 0.1s
=> [internal] load build context                                  0.3s
=> => transferring context: 79.10kB                                0.3s
=> [2/5] WORKDIR /app/chatService                               0.2s
=> [3/5] COPY chatService/package*.json ./                      0.1s
=> [4/5] RUN npm install --production                           15.1s
=> [5/5] COPY chatService/. ./                                  0.2s
=> exporting to image                                           4.6s
=> => exporting layers                                              2.3s
=> => exporting manifest sha256:339aeala6bb78ff1ee3741ed041ce9240d21007bb95e34089dc20321caf3d55d 0.0s
=> => exporting config sha256:d2c417fc0f5883505e79d9e176e9f99cd5fae67d0a9df57ff49f76596a2e9d12 0.0s
=> => exporting attestation manifest sha256:b4701d035b7394284128acc5eb235a32a86072f471a974cfe0e67ca348cdc5a1 0.0s
=> => exporting manifest list sha256:95f826875bec0e10757a22c1c4329de5fdc908b4a4c65bacac94598cac6bb7b3 0.0s
=> => naming to docker.io/harshrajsinhparmar/streaming-chat:1.0.0 0.0s
=> => unpacking to docker.io/harshrajsinhparmar/streaming-chat:1.0.0 2.0s

View build details: docker-desktop://dashboard/build/desktop-linux/desktop-linux/hpzy5xhnmf2oxo7p6ru9hechf
```

5)

```
C:\Users\Hirensinh\Downloads\Container Orchestration Assignment\StreamingApp>docker build -t harshrajsinhparmar/streaming-frontend:1.0.0 ./frontend
2026/09/15 12:26:11 http2: server: error reading preface from client //./pipe/dockerDesktopLinuxEngine: file has already[+] Building 935.7s (14/15)
=> [auth] library/nginx:pull token for registry-1.docker.io                                0.0s
=> [internal] load .dockerignore                                                            0.1s
=> => transferring context: 93B                                                            0.0s
=> [build 1/6] FROM docker.io/library/node:18-alpine@sha256:8d6421d663b4c28fd3ebc498332f249011d118945588d0a35cb9 0.2s
=> => resolve docker.io/library/node:18-alpine@sha256:8d6421d663b4c28fd3ebc498332f249011d118945588d0a35cb9bc4b8c 0.2s
=> [internal] load build context                                                            5.4s
=> => transferring context: 947.16kB                                                       5.2s
=> [production 1/2] FROM docker.io/library/nginx:1.27-alpine@sha256:65645c7bb6a0661892a8b03b89d0743208a18dd2f3f 15.2s
=> => resolve docker.io/library/nginx:1.27-alpine@sha256:65645c7bb6a0661892a8b03b89d0743208a18dd2f3f17a54ef4b76f 0.2s
=> sha256:39c2ddfd6010082a4a646e7ca44e95aca9bf3eaebc00f17f7ccc29540004f2a7d 15.52MB / 15.52MB 12.3s
=> sha256:197eb75867ef4fccc4d724f17b0972ab0489436860a594a9445f8eaff8155053 1.21kB / 1.21kB 1.3s
=> sha256:34a64644b756511a2e217f0508e11d1a572085d66cd6dc9a555a082ad49a3102 1.40kB / 1.40kB 1.3s
=> sha256:81bd8ed7ec6789b0cb7f1b47ee731c522f6dba83201ec73cd6bca1350f582948 402B / 402B 1.1s
=> sha256:d7e5070240863957eb0b5a44a5729963c3462666baa2947d00628cb5f2d5773 955B / 955B 0.6s
=> sha256:b464cfd42a6319875aeb27359ec549790ce14d8214fcb16ef915e4530e5ed235 629B / 629B 0.6s
=> sha256:61ca4f733c802af9e05a32f0de0361b6d713b8b53292dc15fb093229f648674 1.79MB / 1.79MB 6.5s
=> extracting sha256:61ca4f733c802af9e05a32f0de0361b6d713b8b53292dc15fb093229f648674 0.9s
=> extracting sha256:b464cfd42a6319875aeb27359ec549790ce14d8214fcb16ef915e4530e5ed235 0.1s
=> extracting sha256:d7e5070240863957eb0b5a44a5729963c3462666baa2947d00628cb5f2d5773 0.1s
=> extracting sha256:81bd8ed7ec6789b0cb7f1b47ee731c522f6dba83201ec73cd6bca1350f582948 0.1s
=> extracting sha256:197eb75867ef4fccc4d724f17b0972ab0489436860a594a9445f8eaff8155053 0.1s
=> extracting sha256:34a64644b756511a2e217f0508e11d1a572085d66cd6dc9a555a082ad49a3102 0.1s
=> extracting sha256:39c2ddfd6010082a4a646e7ca44e95aca9bf3eaebc00f17f7ccc29540004f2a7d 1.7s
=> CACHED [build 2/6] WORKDIR /app                                                         0.0s
=> [build 3/6] COPY package*.json ./                                                       0.2s
=> [build 4/6] RUN npm install                                                            410.7s
=> [build 5/6] COPY . .                                                                  8.0s
=> CANCELED [build 6/6] RUN npm run build                                                 507.2s
```

Stop Minikube to free 100% of RAM for Docker

```
C:\Users\Hirensinh\Downloads\Container Orchestration Assignment\StreamingApp>docker build -t harshrajsinhparmar/streaming-frontend:1.0.0 ./frontend
[+] Building 832.9s (14/15)                                                                 docker:desktop-linux
=> [internal] load build definition from Dockerfile                                         0.1s
=> => transferring dockerfile: 846B                                                         0.1s
=> [internal] load metadata for docker.io/library/nginx:1.27-alpine                       2.6s
=> [internal] load metadata for docker.io/library/node:18-alpine                         2.6s
=> [auth] library/nginx:pull token for registry-1.docker.io                             0.0s
=> [auth] library/node:pull token for registry-1.docker.io                             0.0s
=> [internal] load .dockerignore                                                            0.1s
=> => transferring context: 93B                                                            0.0s
=> CACHED [production 1/2] FROM docker.io/library/nginx:1.27-alpine@sha256:65645c7bb6a0661892a8b03b89d0743208a18dd2f3f17a54ef4b76fb8 0.2s
=> => resolve docker.io/library/nginx:1.27-alpine@sha256:65645c7bb6a0661892a8b03b89d0743208a18dd2f3f17a54ef4b76fb8e2f2a10 0.2s
=> [build 1/6] FROM docker.io/library/node:18-alpine@sha256:8d6421d663b4c28fd3ebc498332f249011d118945588d0a35cb9bc4b8ca09d9e 0.3s
=> => resolve docker.io/library/node:18-alpine@sha256:8d6421d663b4c28fd3ebc498332f249011d118945588d0a35cb9bc4b8ca09d9e 0.3s
=> [internal] load build context                                                            0.1s
=> => transferring context: 2.61kB                                                         0.1s
=> CACHED [build 2/6] WORKDIR /app                                                         0.0s
=> CACHED [build 3/6] COPY package*.json ./                                               0.0s
=> CACHED [build 4/6] RUN npm install                                                      0.0s
=> CACHED [build 5/6] COPY . .                                                            0.0s
=> CANCELED [build 6/6] RUN npm run build                                                 829.3s

2 warnings found (use docker --debug to expand):
- SecretsUsedInArgOrEnv: Do not use ARG or ENV instructions for sensitive data (ARG "REACT_APP_AUTH_API_URL") (line 10)
- SecretsUsedInArgOrEnv: Do not use ARG or ENV instructions for sensitive data (ENV "REACT_APP_AUTH_API_URL") (line 17)
ERROR: failed to build: failed to solve: Canceled: context canceled

C:\Users\Hirensinh\Downloads\Container Orchestration Assignment\StreamingApp>minikube stop
* Stopping node "minikube" ...
* Powering off "minikube" via SSH ...
* 1 node stopped.
```

```
C:\Users\Hirensinh\Downloads\Container Orchestration Assignment\StreamingApp>docker build -t harshrajsinhparmar/streaming-frontend:1.0.0 ./frontend
[*] Building 219.4s (16/16) FINISHED
=> [internal] load build definition from Dockerfile
=> => transferring dockerfile: 846B
=> [internal] load metadata for docker.io/library/node:18-alpine
=> [internal] load metadata for docker.io/library/nginx:1.27-alpine
=> [auth] library/node:pull token for registry-1.docker.io
=> [auth] library/nginx:pull token for registry-1.docker.io
=> [internal] load .dockerignore
=> => transferring context: 93B
=> [build 1/6] FROM docker.io/library/node:18-alpine@sha256:8d6421d663b4c28fd3ebc498332f249011d118945588d0a35cb9bc4b8ca09d9e
=> => resolve docker.io/library/node:18-alpine@sha256:8d6421d663b4c28fd3ebc498332f249011d118945588d0a35cb9bc4b8ca09d9e
=> CACHED [production 1/2] FROM docker.io/library/nginx:1.27-alpine@sha256:65645c7bb6a0661892a8b03b89d0743208a18dd2f3f17a54ef4b76fb8
=> => resolve docker.io/library/nginx:1.27-alpine@sha256:65645c7bb6a0661892a8b03b89d0743208a18dd2f3f17a54ef4b76fb8e2f2a10
=> [internal] load build context
=> => transferring context: 2.61kB
=> CACHED [build 2/6] WORKDIR /app
=> CACHED [build 3/6] COPY package*.json ./
=> CACHED [build 4/6] RUN npm install
=> CACHED [build 5/6] COPY . .
=> [build 6/6] RUN npm run build
=> [production 2/2] COPY --from=build /app/build /usr/share/nginx/html
=> exporting to image
=> => exporting layers
=> => exporting manifest sha256:d098a827c167783249d8599cae8d5174093f7029a6a204b3e15866e057c17090
=> => exporting config sha256:ce09722eefa15974b6a5e0417e4623ca6baf2959d41bf1a4669aac658576c759
=> => exporting attestation manifest sha256:9808e9aa60d60102e737d31c4737814205a06f1580bd87df7b3cd30127ad9822
=> => exporting manifest list sha256:41ebd8daf304b0d101bcf4969fbbc27634d59bffb0c0ec45d27553ff961f5f95
=> => naming to docker.io/harshrajsinhparmar/streaming-frontend:1.0.0
=> => unpacking to docker.io/harshrajsinhparmar/streaming-frontend:1.0.0

2 warnings found (use docker --debug to expand):
- SecretsUsedInArgOrEnv: Do not use ARG or ENV instructions for sensitive data (ARG "REACT_APP_AUTH_API_URL") (line 10)
- SecretsUsedInArgOrEnv: Do not use ARG or ENV instructions for sensitive data (ENV "REACT_APP_AUTH_API_URL") (line 17)
```

C:\Users\Hirensinh\Downloads\Container Orchestration Assignment\StreamingApp>docker images harshrajsinhparmar/*

IMAGE	ID	DISK USAGE	CONTENT SIZE	EXTRA
harshrajsinhparmar/notes-app-k8:latest	d81853dff74c	1.69GB	430MB	
harshrajsinhparmar/streaming-admin:1.0.0	45be2235ac99	272MB	59.6MB	
harshrajsinhparmar/streaming-auth:1.0.0	532dee24430d	241MB	56.4MB	
harshrajsinhparmar/streaming-chat:1.0.0	95f826875bec	230MB	54.8MB	
harshrajsinhparmar/streaming-frontend:1.0.0	41ebd8daf304	78.2MB	21.9MB	
harshrajsinhparmar/streaming-stream:1.0.0	d3dabb86d7d5	303MB	74.9MB	

Push All 5 Images to Docker Hub

```
C:\Users\Hirensinh\Downloads\Container Orchestration Assignment\StreamingApp>docker push harshrajsinhparmar/streaming-frontend:1.0.0
The push refers to repository [docker.io/harshrajsinhparmar/streaming-frontend]
821799a8afab: Pushed
677403a47155: Pushed
81bd8ed7ec67: Pushed
34a64644b756: Pushed
61ca4f733c80: Pushed
b464cdf2a63: Pushed
d7e507024086: Pushed
197eb75867ef: Pushed
f18232174bc9: Pushed
39c2ddf6010: Pushed
1.0.0: digest: sha256:41ebd8daf304b0d101bcf4969fbbc27634d59bffb0c0ec45d27553ff961f5f95 size: 856
```

```
C:\Users\Hirensinh\Downloads\Container Orchestration Assignment\StreamingApp>docker push harshrajsinhparmar/streaming-chat:1.0.0
The push refers to repository [docker.io/harshrajsinhparmar/streaming-chat]
b5398f0828d9: Pushed
b579db047a8a: Pushed
e381e4b85d15: Pushed
dd71dde834b5: Mounted from harshrajsinhparmar/streaming-stream
1e5a4c89cee5: Mounted from harshrajsinhparmar/streaming-stream
f18232174bc9: Mounted from harshrajsinhparmar/streaming-stream
14d90a175251: Pushed
d22d0e92191b: Pushed
25ff2da83641: Mounted from harshrajsinhparmar/streaming-stream
1.0.0: digest: sha256:95f826875bec0e10757a22c1c4329de5fdc908b4a4c65bacac94598cac6bb7b3 size: 856
```

```

C:\Users\Hirensinh\Downloads\Container Orchestration Assignment\StreamingApp>docker push harshrajsinhparmar/streaming-auth:1.0.0
The push refers to repository [docker.io/harshrajsinhparmar/streaming-auth]
55abce9ff6e3: Pushed
44b0141e8d78: Pushed
dd71dde834b5: Pushed
619c83c69533: Pushed
1e5a4c89cee5: Pushed
f18232174bc9: Mounted from harshrajsinhparmar/streaming-frontend
25ff2da83641: Pushed
a80aa6787550: Pushed
59305a0c8411: Pushed
1.0.0: digest: sha256:532dee24430d7a4200e695677137d7468df448da64ad2631502956725f1cd729 size: 856

C:\Users\Hirensinh\Downloads\Container Orchestration Assignment\StreamingApp>docker push harshrajsinhparmar/streaming-stream:1.0.0
The push refers to repository [docker.io/harshrajsinhparmar/streaming-stream]
82504408674a: Pushed
ac973cd93355e: Pushed
dd71dde834b5: Mounted from harshrajsinhparmar/streaming-auth
25ff2da83641: Mounted from harshrajsinhparmar/streaming-auth
f18232174bc9: Mounted from harshrajsinhparmar/streaming-frontend
8e084568375a: Pushed
42098cbf664d: Pushed
1e5a4c89cee5: Mounted from harshrajsinhparmar/streaming-auth
ac4b4b753101: Pushed
1.0.0: digest: sha256:d3dabb86d7d58ac5dbb18423b15c2d267580ddd47d759446d5eb2c6def10135d size: 856

C:\Users\Hirensinh\Downloads\Container Orchestration Assignment\StreamingApp>docker push harshrajsinhparmar/streaming-admin:1.0.0
The push refers to repository [docker.io/harshrajsinhparmar/streaming-admin]
d8f667e3b515: Pushed
25ff2da83641: Mounted from harshrajsinhparmar/streaming-stream
dd71dde834b5: Mounted from harshrajsinhparmar/streaming-stream
910fbe471d03: Pushed
f18232174bc9: Mounted from harshrajsinhparmar/streaming-stream
7b4080af17ef: Pushed
16954a11d720: Pushed
1e5a4c89cee5: Mounted from harshrajsinhparmar/streaming-stream
c2fb816f7b22: Pushed
1.0.0: digest: sha256:45be2235ac996292eaff3e3440d513e106aa529024526021a47belebabeabf3f5 size: 856

```

Restart Minikube if it is still stopped

```

C:\Users\Hirensinh\Downloads\Container Orchestration Assignment\StreamingApp>minikube start
* minikube v1.37.0 on Microsoft Windows 11 Pro 10.0.26200.9445 Build 26200.9445
* Using the docker driver based on existing profile
* Starting "minikube" primary control-plane node in "minikube" cluster
* Pulling base image v0.0.48 ...
* Restarting existing docker container for "minikube" ...
! Failing to connect to https://registry.k8s.io/ from inside the minikube container
* To pull new external images, you may need to configure a proxy: https://minikube.sigs.k8s.io/docs/reference/networking/proxy/
* Preparing Kubernetes v1.34.0 on Docker 28.4.0 ...
* Verifying Kubernetes components...
  - Using image gcr.io/k8s-minikube/storage-provisioner:v5
* After the addon is enabled, please run "minikube tunnel" and your ingress resources would be available at "127.0.0.1"
  - Using image registry.k8s.io/ingress-nginx/controller:v1.13.2
  - Using image registry.k8s.io/ingress-nginx/kube-webhook-certgen:v1.6.2
  - Using image registry.k8s.io/ingress-nginx/kube-webhook-certgen:v1.6.2
* Verifying ingress addon...
* Enabled addons: default-storageclass, storage-provisioner, ingress
* Done! kubectl is now configured to use "minikube" cluster and "default" namespace by default

```